AP Access & Course-Taking Equity

A reproducible district-facing analysis of 2023–24 CRDC advanced-course data linked to the NCES Common Core of Data

Research memo · Independent portfolio project · October 2026

GRADE-12 SCHOOLS	AP ACCESS	REPORTED AP PARTICIPANTS
24,872	**53.7%**	**3,016,886**
Primary analytic universe	95% CI 53.0%–54.3%	18.1% of enrollment

Executive summary

Among 24,872 schools offering grade 12, 53.7% reported at least one AP course. The estimate is precise nationally, but the dashboard exposes wide state and district variation and warns against interpreting small-district percentages without their confidence intervals.

This study is observational. Its regression estimates associations—not effects of school composition—and its race/ethnicity representation measure compares shares among reported AP participants with shares of total school enrollment rather than establishing equal opportunity or causal mechanisms.

Recommended use

Use the district explorer to identify where AP course availability warrants follow-up; then pair the signal with course catalogs, staffing, prerequisites, student demand, and alternative advanced-course pathways before making program decisions.

State variation

State-level access estimates vary substantially in the selected universe. These contrasts are useful for generating questions, but they also reflect different school structures, course-reporting practices, virtual options, and advanced-course alternatives.

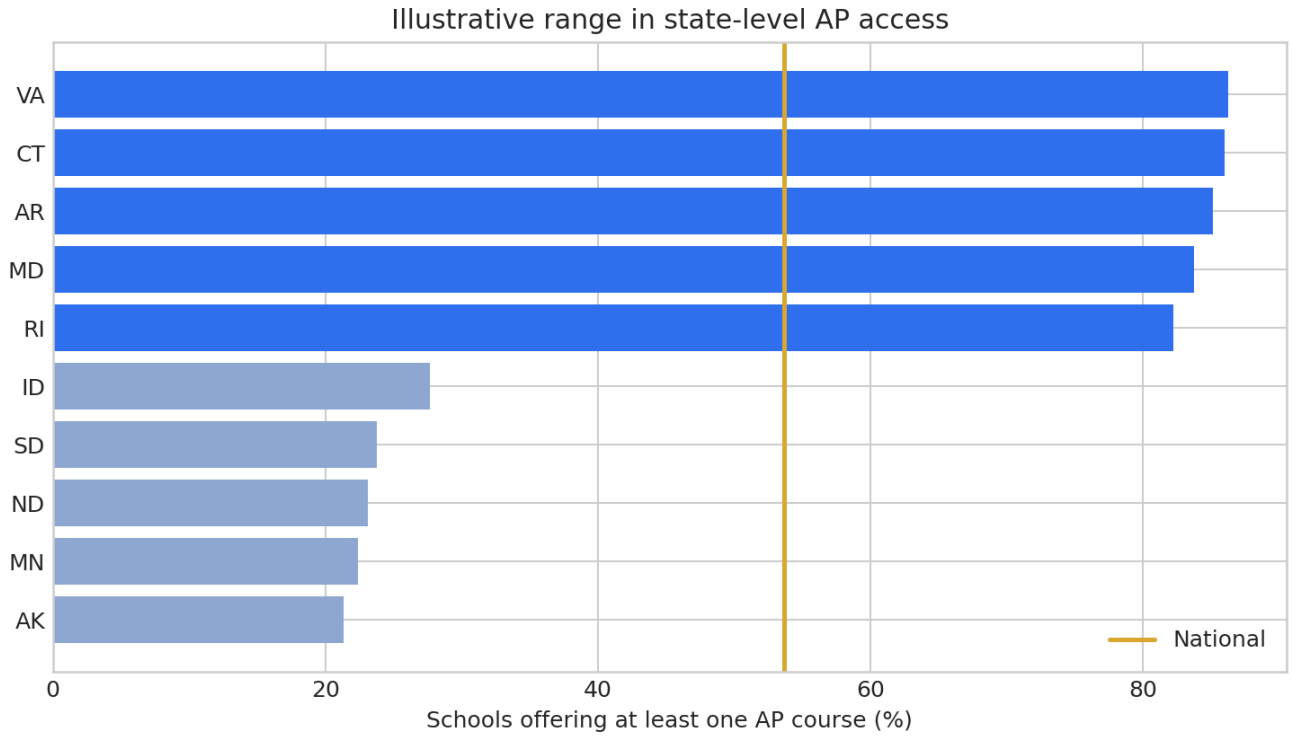


Figure 1. Five highest and five lowest state/jurisdiction access estimates in this analytic universe. State context and reporting practices may differ.

Equity signals

The representation-gap view asks whether each reported racial or ethnic group comprises a larger or smaller share of AP participants than of total enrollment across the included schools. Negative values identify a descriptive disparity that merits local investigation; they do not identify its cause.

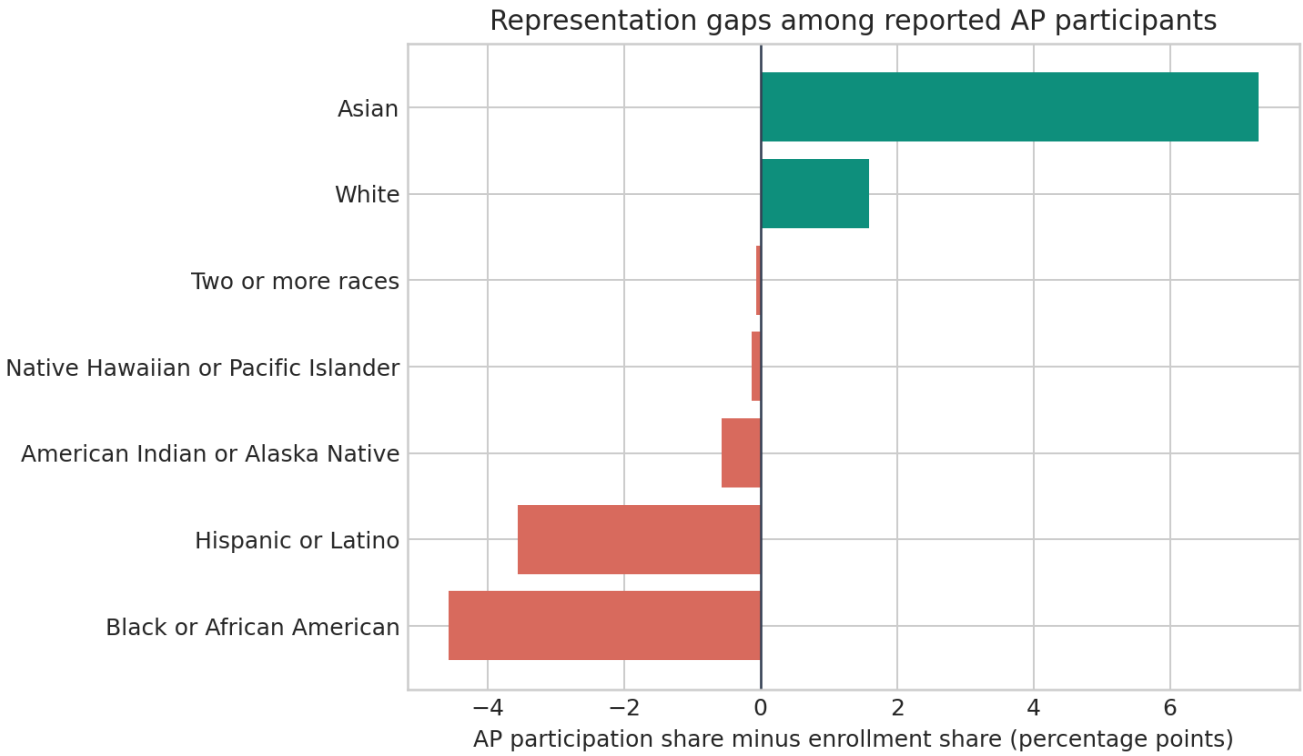


Figure 2. AP participation share minus enrollment share. Counts use nonnegative reported sex-component values; suppressed or unavailable components can understate totals.

What stands out

• **Black or African American:** -4.6 pp; 2,448,292 enrolled students and 304,690 reported AP participants.

• **Hispanic or Latino:** -3.6 pp; 4,883,136 enrolled students and 775,751 reported AP participants.

• **Asian:** +7.3 pp; 858,221 enrolled students and 375,534 reported AP participants.

Questions for district leaders

Which schools do not offer AP? Which subjects are absent? How do prerequisites, counseling, scheduling, transportation, teacher availability, and alternative advanced courses shape access and participation? Are results stable when examined by grade, disability status, English learner status, and school size?

Guardrail

The dashboard is a screening and inquiry tool—not a school ranking. Any high-stakes interpretation should return to source records and engage district staff who understand local course structures.

Model and sensitivity analysis

A school-level logistic regression estimates the odds of offering at least one AP course. Continuous predictors are standardized; uncertainty uses 150 bootstrap replications clustered by state. The complete-case model contains 17,989 schools.

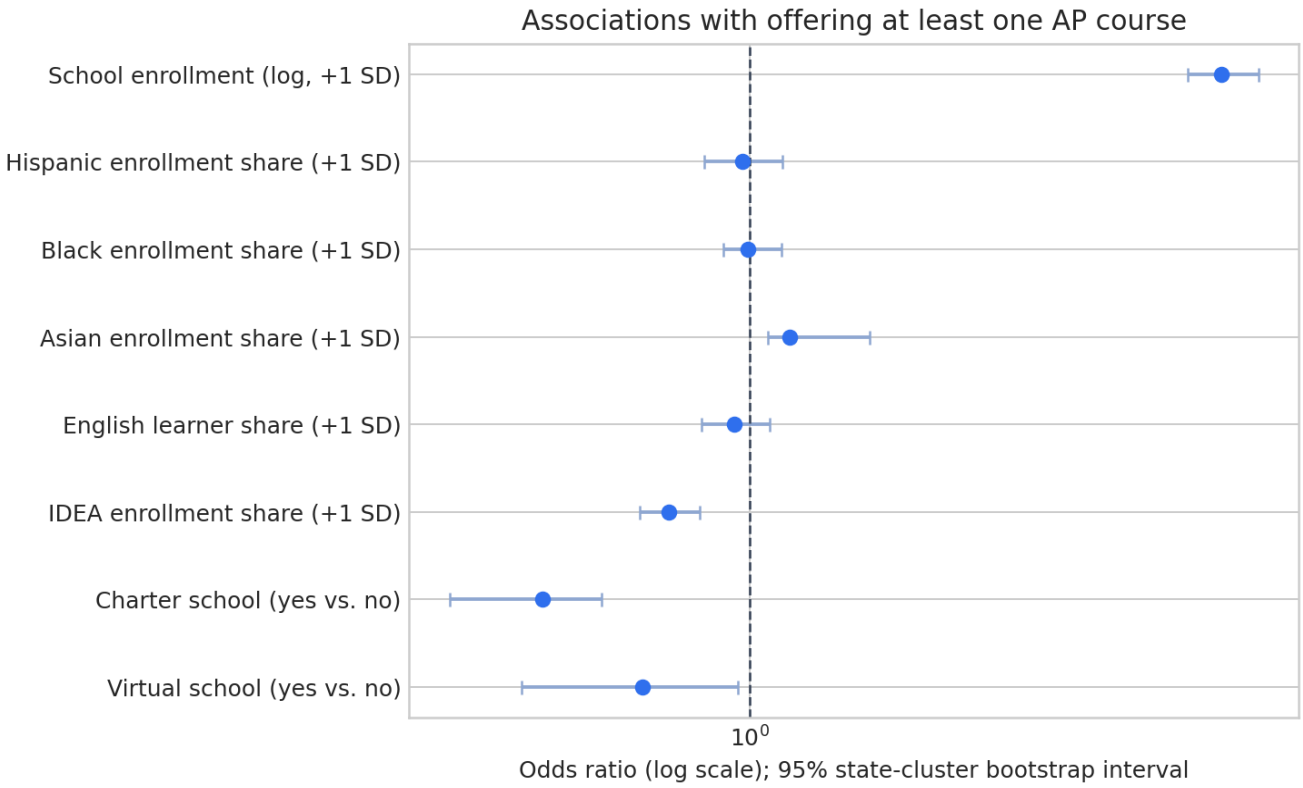


Figure 3. Odds ratios and 95% state-cluster bootstrap intervals. Estimates are descriptive associations and can reflect unmeasured school, district, and state factors.

Sensitivity checks

Scenario	Schools	Access	95% CI
Primary: grade 12 offered	24,871	53.7%	53.0%–54.3%
CCD school level = High	21,159	58.8%	58.2%–59.5%
Enrollment at least 100	19,942	65.3%	64.6%–66.0%
Alternative outcome: at least one AP participant	24,871	53.5%	52.8%–54.1%

The national conclusion depends on the analytic universe: restricting to CCD schools classified as high schools raises access to 58.8%, and requiring at least 100 students raises it to 65.3%. The dashboard therefore keeps the universe definition visible beside every estimate.

Data pipeline and quality controls

Source files

2023–24 Civil Rights Data Collection Advanced Placement, Enrollment, and School Characteristics files; 2023–24 NCES Common Core of Data Public School Directory.

Stage	Records	Rule / result
CRDC AP file	97,094	Unique COMBOKEY school records
CRDC file linkage	97,094	100% matched to enrollment and characteristics
CCD linkage	94,382	97.2% matched; 2,712 unmatched
Grade-12 universe	24,872	CCD G_12_OFFERED = Yes
Known AP access	24,871	1 special-code record excluded
Complete-case model	17,989	Nonmissing outcome and covariates

Identifiers and joins

CRDC COMBOKEY and CCD NCESSCH were standardized as zero-padded 12-character strings. CRDC files were validated as one-to-one joins. The combined CRDC data were left-joined to CCD so unmatched CRDC records remained auditable.

Missingness and reserve codes

Negative CRDC reserve codes are never treated as student counts. For grade-12 schools, SCH_APCOURSES = –9 (not applicable/skipped) is interpreted as no AP offering; –5 and other exceptional codes remain missing. Sex-component totals sum available nonnegative values, which can understate totals when a component is unavailable or suppressed.

Reproducibility package

The project publishes the cleaned school file, state and district summaries, model and sensitivity tables, data dictionary, pipeline documentation, and the Python script that recreates the outputs from the four official source files.

Independence and limitations

This independent portfolio project is not affiliated with or endorsed by the College Board, the U.S. Department of Education, or NCES. AP is a registered program name of the College Board. Source agencies remain authoritative for their data.

Recommended action framework

1 · Validate the signal

Confirm the district and school identifiers, course-offering definitions, suppression codes, and confidence intervals. Compare the dashboard with local transcripts, course catalogs, and scheduling data.

2 · Diagnose the opportunity

Disaggregate availability and participation, then investigate student demand, prerequisites, counselor recommendations, staffing, scheduling conflicts, transportation, and awareness. Include other advanced-course routes so AP is not treated as the only path.

3 · Co-design a response

Work with district leaders, school staff, students, and families to define the decision, feasible intervention, owner, timeline, and equity guardrails before selecting a metric.

4 · Monitor and learn

Track leading measures—course sections, seats, requests, counselor outreach, prerequisite completion—alongside access and participation. Predefine review dates and a stop/adjust/scale rule.

Suggested follow-up study

Build a longitudinal student-level file linking grade 8–12 course taking, school context, AP participation and performance, graduation, and postsecondary enrollment. Use district-specific questions, documented linkage rates, subgroup privacy rules, and a model appropriate to students nested within schools and districts.

Interpretive checklist

- ■ Is the question about course availability, student enrollment, exam taking, or longer-term outcomes?
- ■ Are denominators and missing-value rules visible?
- ■ Are small-n estimates shown with uncertainty and privacy protections?
- ■ Are observed disparities described without implying causality?
- ■ Can a district partner reproduce the result from the documentation?